

Ultrasmall 8-Channel Wavelength/Mode Hybrid Multiplexer with Mosaic Structures for Beyond-Tbit/s Optical Ethernet

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Abstract We demonstrate ultrasmall wavelength/mode hybrid multiplexers for the C- and O-bands. Integrating mosaic structures designed using machine learning enables ultrasmall footprint of $50 \times 50 \mu\text{m}^2$ while maintaining low loss property. To the best of our knowledge, these devices are the smallest for 8-channel multiplexers. ©2026 The Author(s)

Introduction

Recently, rapid progress in artificial intelligence is driving the development of high-capacity and power efficient optical interconnect technologies. The transmission capacity of optical communication systems has expanded with increasing signal speeds and the use of wavelength division multiplexing (WDM). For example, coarse WDM (CWDM) systems have been employed in several Ethernet standards, such as 400GBASE-FR4 [1]. The key device for WDM systems is a wavelength multiplexer (MUX), and various wavelength MUXes have been proposed [2, 3]. In addition, polarisation division multiplexing (PDM) and mode division multiplexing (MDM) have also been studied as multichannel techniques. To further increase the transmission capacity, combining PDM or MDM with WDM is considered as a promising approach, and various wavelength/mode hybrid MUXes have been proposed [4–6]. In [6], a 4λ , 2-mode MUX based on a combination of WDM and MDM was proposed. It enables the 8-channel operation without polarisation handling devices which require structural asymmetry in the thickness direction to achieve polarisation rotation. However, the 4λ , 2-mode hybrid MUX still has a large footprint of $600 \times$

$1000 \mu\text{m}^2$.

This work focuses on the drastic downsizing of the 4λ , 2-mode hybrid MUX, aiming to reduce its size to 1/100 of that of the previously reported device. To achieve drastic downsizing, we utilize mosaic structures, which enable the extreme downsizing of passive devices [7–11]. In mosaic structures, nanoholes are placed in a silicon plate to obtain desired passive functions. In our previous studies, we have reported mosaic-based power splitters [8,9], a mode MUX [10], and a wavelength MUX [11], all of which were designed using machine learning, demonstrating the usefulness of mosaic structures.

In this paper, we demonstrate ultrasmall 4λ , 2-mode hybrid MUXes for the C- and O-bands, respectively. The schematic of the proposed hybrid MUX is shown in Fig. 1. Two mosaic-based 4λ MUXes and one mosaic-based 2-mode MUX are integrated within a small footprint of $50 \times 50 \mu\text{m}^2$. The channel spacing of the 4λ MUXes is 20 nm, which is compatible with CWDM systems. Each mosaic structure is designed by Bayesian direct binary search (DBS) method, in which Bayesian optimisation assists a DBS algorithm. After demonstrating the C-band 8-channel hybrid MUX

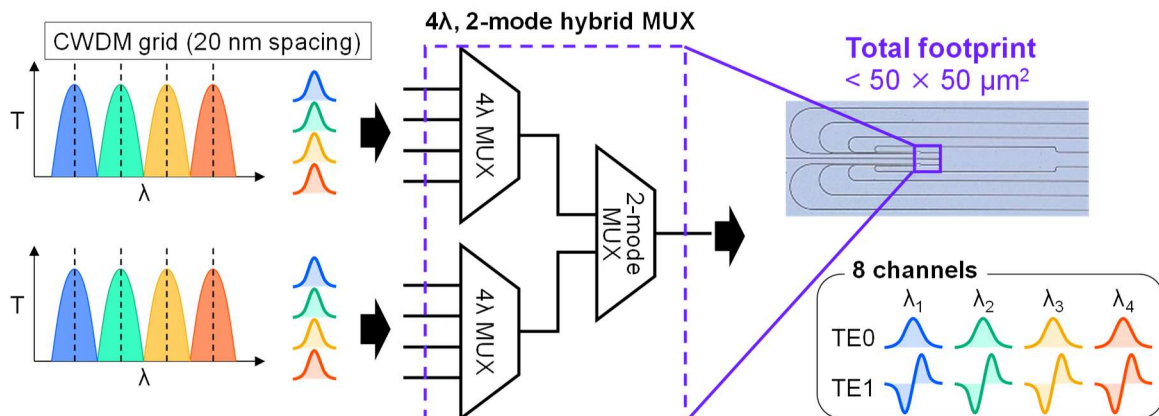


Fig. 1: Schematic of 8-channel wavelength/mode hybrid MUX.

as a proof of concept, we present the results for the O-band 8-channel hybrid MUX. To the best of our knowledge, this is the smallest footprint reported for 8-channel MUXes in the C- and O-bands, leading to the ultracompact high-capacity transmitter chips based on WDM and MDM technologies, while maintaining low-loss performance.

Design of Mosaic Device

The basic algorithm for the device design is a direct binary search (DBS) method. The method is often used in this field [7–11]. A silicon plate is discretized into small pixels, in which a nanohole is placed depending on the design. A normal DBS method involves the following steps:

1. Selecting a pixel in a random order,
2. changing the pixel's state (Si or nanohole),
3. simulating and evaluating device performance,
4. and judging whether to accept the change in the pixel's state.

The judgement is made to minimise a figure of merit (FOM), which indicates the device performance.

We also used Bayesian DBS method in this work. Calculated data that are obtained through a DBS method are accumulated, and Bayesian optimisation assists the step “selecting a pixel” in the DBS algorithm by exploiting the accumulated data, improving optimisation efficiency. A detailed explanation of this method and the efficient design of power splitters have been presented in [8].

Fig. 2(a) shows the schematic of the mosaic-based 2-mode MUX for the C-band. The

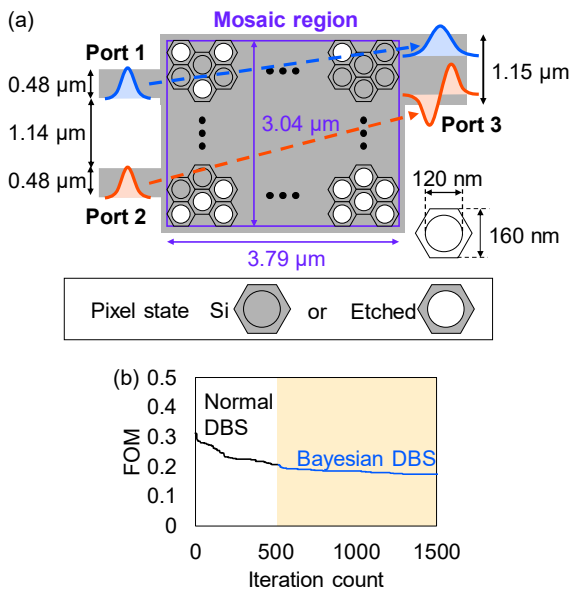


Fig. 2: (a) Schematic and (b) improvement of FOM as a function of iteration count for C-band mosaic-based 2-mode MUX.

dimensions of the mosaic region are $3.04 \mu\text{m} \times 3.79 \mu\text{m}$. Hexagonal pixels are arranged on a triangular lattice, and the number of pixels is 500. When a triangular lattice is employed, a mosaic region can be utilized more efficiently since its filling ratio is larger than that of a square lattice [10,11]. The lattice constant and hole diameter are 160 and 120 nm, respectively. The widths of input and output waveguides are 0.48 and $1.15 \mu\text{m}$, respectively. The FOM is defined as follows.

$$\text{FOM} = \frac{1}{N} \sum_{i=1}^N \{ (1 - \alpha)(1 - T_{31, \text{TE0}}) + \alpha(1 - T_{32, \text{TE1}}) \} \quad (1)$$

$T_{31, \text{TE0}}$ and $T_{32, \text{TE1}}$ denote the coupling efficiencies to TE0 and TE1 modes obtained at port 3 when the TE0 mode is launched from port 1 and port 2, respectively. α is the weight coefficient, and we used $\alpha = 0.6$. The improvement of the FOM as a function of iteration count for the 2-mode MUX design is shown in Fig. 2(b). Bayesian DBS method was used after 500 iterations of the normal DBS method.

For the C-band 4λ MUX, we utilize the mosaic structure previously reported in [11]. The 4λ MUX was also designed by Bayesian DBS method. The dimensions of the mosaic region are $7.58 \mu\text{m} \times 13.8 \mu\text{m}$.

Results and Discussion

Fig. 3(a) shows the scanning electron microscope (SEM) image of the fabricated 8-channel hybrid (de)MUX illustrated in Fig. 1. Figs. 3(b) and 3(c) show the enlarged views of the 2-mode (de)MUX and 4λ (de)MUX, respectively. The mosaic devices are connected by bends with a radius of $10 \mu\text{m}$. Electron-beam lithography and dry etching process were used to form the mosaic structure, and the entire device is covered with a SiO_2 upper cladding. Measurement was carried out under demultiplexing operation.

Fig. 4(a) shows the calculated and measured

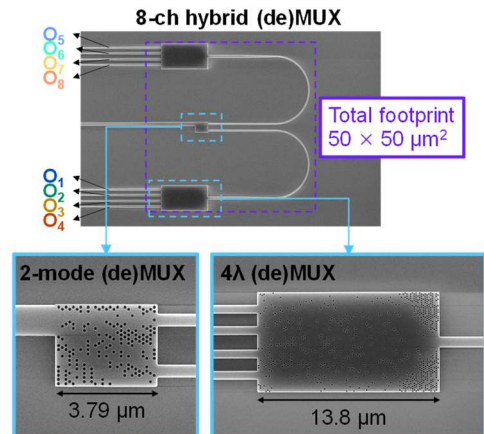


Fig. 3: SEM images of (a) 8-channel hybrid (de)MUX, and enlarged views of (b) 2-mode (de)MUX, and (c) 4λ (de)MUX.

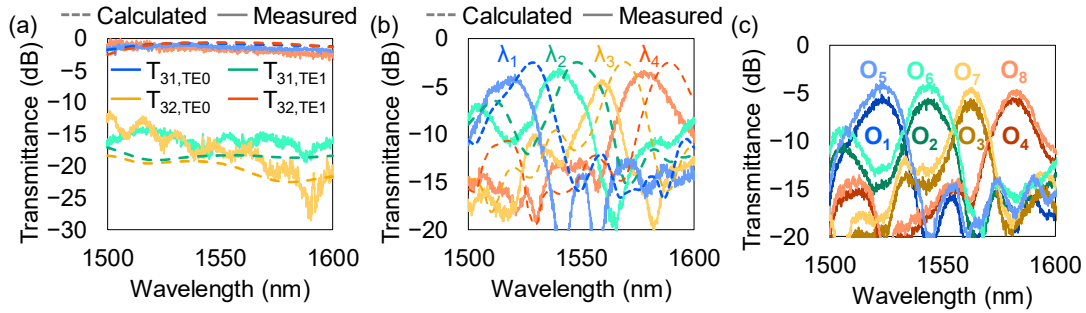


Fig. 4: Calculated and measured transmittance spectra of C-band (a) 2-mode (de)MUX and (b) 4λ (de)MUX, and measured transmittance spectra of C-band 8-channel hybrid (de)MUX.

transmittance spectra of the C-band 2-mode (de)MUX. $T_{31,TE1}$ and $T_{32,TE0}$ represent the modal crosstalk. The mosaic-based 2-mode (de)MUX exhibits the broadband operation. The insertion loss is lower than 2 dB over the C-band. Fig. 4(b) shows the calculated and measured transmittance spectra of the C-band 4λ (de)MUX. Although the measured spectra are blue-shifted from calculated spectra, the clear 4λ operation is achieved. The insertion loss at the peak wavelengths is about 4 dB. Fig. 4 (c) shows the measured transmittance spectra of the C-band 8-channel hybrid (de)MUX. The positions of output ports O_1 – O_8 are illustrated in Fig. 3. The two groups of 4λ operation are clearly achieved with insertion losses of 4–6 dB. The insertion losses for O_1 – O_4 are about 1 dB higher than those for O_5 – O_8 . The additional loss is caused by the mode conversion from the TE1 mode to the TE0 mode. The 4λ operation for the TE0 and the TE1 modes is achieved within an area only $50 \times 50 \mu\text{m}^2$. The proposed device occupies only 1/240 of the footprint of the device in [6], and the insertion loss is about 3 dB lower than that of the device in [6].

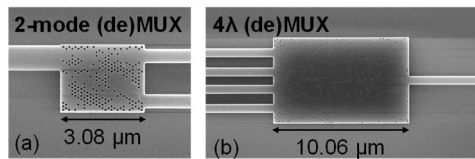


Fig. 5: SEM images of O-band (a) 2-mode (de)MUX and (b) 4λ (de)MUX.

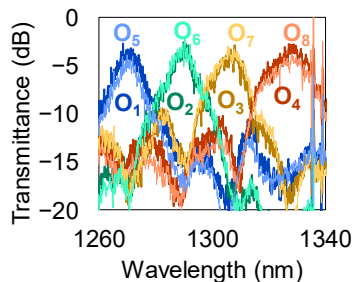


Fig. 6: Measured transmittance spectra of O-band 8-channel hybrid (de)MUX.

We also demonstrate the O-band 8-channel hybrid (de)MUX. The device configuration is the same as the C-band one. The 2-mode (de)MUX and 4λ (de)MUX were newly designed for the O-band. Each device was designed using the same optimisation method, but the smaller values of 130 and 90 nm are used for the lattice constant and hole diameter, respectively, because smaller pixels are preferable for shorter wavelengths.

Figs. 5(a) and 5(b) show the SEM images of the O-band 2-mode (de)MUX and 4λ (de)MUX. The dimensions of the mosaic regions of the 2-mode (de)MUX and 4λ (de)MUX are $2.47 \mu\text{m} \times 3.08 \mu\text{m}$ and $6.37 \mu\text{m} \times 10.06 \mu\text{m}$, respectively. The insertion loss of the O-band (de)MUX is about 1 dB lower than that presented in [11]. Fig. 6 shows the measured transmittance spectra of the O-band 8-channel hybrid (de)MUX. The device exhibits the clear 8-channel operation with low insertion losses of 3–4 dB.

The use of higher-order modes in MDM contributes to the scalability of multichannel techniques, along with WDM. In addition, by integrating a TE1-TM0 mode converter [12] with the proposed 8-channel hybrid MUX, it can be used as a dual-polarisation, 4λ MUX. In combination with 200-Gbit/s PAM4 transmitters, this configuration can achieve terabit/s optical Ethernet.

Conclusions

We have proposed the 4λ, 2-mode hybrid MUXes for the C- and O-bands, respectively. The usefulness of an integrated device composed of multiple mosaic structures is demonstrated, and the O-band 8-channel MUX is very promising as a passive device for the multi-terabit/s optical Ethernet. By using mosaic devices, the footprint of Si-photonics transmitter chips becomes negligible while maintaining low optical loss, showing the suitability for multifunctional photonic integrated circuits that meet demands for lower cost and lower power consumption.

References

- [1] Ethernet alliance, "2026 Ethernet roadmap", [Online] Available: <https://ethernetalliance.org/>
- [2] S. Chen, X. Fu, J. Wang, Y. Shi, S. He, and D. Dai, "Compact dense wavelength-division (de)multiplexer utilizing a bidirectional arrayed-waveguide grating integrated with a Mach-Zehnder interferometer", *Journal of Lightwave Technology*, vol. 33, no. 11, pp. 2279–2285, Jun. 2015, DOI: [10.1109/JLT.2015.2405510](https://doi.org/10.1109/JLT.2015.2405510).
- [3] H. Xu and Y. Shi, "Flat-Top CWDM multiplexer based on MZI with bent directional couplers", *IEEE Photonics Technology Letters*, vol. 30, no. 2, pp. 169–172, Jan. 2018, DOI: [10.1109/LPT.2017.2779489](https://doi.org/10.1109/LPT.2017.2779489).
- [4] S. Wang, H. Wu, M. Zhang, and D. Dai, "A 32-channel hybrid wavelength-/mode-division multiplexer on silicon", *IEEE Photonics Technology Letters*, vol. 30, no. 13, pp. 1194–1197, Jul. 2018, DOI: [10.1109/LPT.2018.2839533](https://doi.org/10.1109/LPT.2018.2839533).
- [5] S. Ohta, T. Fujisawa, S. Makino, T. Sakamoto, T. Matsui, K. Tsujikawa, K. Nakajima, and K. Saitoh, "Si-based Mach-Zehnder wavelength/mode multi/demultiplexer for a WDM/MDM transmission system", *Optics Express*, vol. 26, no. 12, pp. 15211–15220, Jun. 2018, DOI: [10.1364/OE.26.015211](https://doi.org/10.1364/OE.26.015211).
- [6] T. Fujisawa and K. Saitoh, "Fabrication-tolerant, 2-mode, 4λ multiplexer based on Si waveguides for beyond Tbit/s optical Ethernet", *2023 Optical Fiber Communications Conference and Exhibition (OFC)*, San Diego, CA, USA, 2023, pp. 1–3, DOI: [10.1364/OFC.2023.M1C.4](https://doi.org/10.1364/OFC.2023.M1C.4).
- [7] B. Shen, P. Wang, R. Polson, and R. Menon, "An integrated-nanophotonics polarization beamsplitter with $2.4 \times 2.4 \mu\text{m}^2$ footprint", *Nature Photonics*, vol. 9, no. 6, pp. 378–382, May 2015. DOI: [10.1038/nphoton.2015.80](https://doi.org/10.1038/nphoton.2015.80).
- [8] T. Fujisawa and K. Saitoh, "Bayesian direct-binary-search algorithm for the efficient design of mosaic-based power splitters", *OSA Continuum*, vol. 4, no. 4, pp. 1258–1270, Apr. 2021. DOI: [10.1364/OSAC.422116](https://doi.org/10.1364/OSAC.422116).
- [9] T. Fujisawa, T. Mitarai, Y. Sawada, T. Okimoto, N. Kono, N. Fujiwara, T. Muratsubaki, and H. Yagi, "Ultrasmall power splitters with triangular lattice mosaic patterns designed by Bayesian DBS method", *IEEE Photonics Technology Letters*, vol. 35, no. 21, pp. 1139–1142, Nov. 2023, DOI: [10.1109/LPT.2023.3306313](https://doi.org/10.1109/LPT.2023.3306313).
- [10] T. Fujisawa, T. Mitarai, T. Okimoto, N. Kono, N. Fujiwara, T. Sato, H. Yagi, and K. Saitoh, "Bayesian design of mosaic-based mode multiplexers for various wavelength bands", *Optics Express*, vol. 31, no. 16, pp. 26842–26853, Jul. 2023, DOI: [10.1364/OE.492673](https://doi.org/10.1364/OE.492673).
- [11] Y. Sawada, T. Fujisawa, T. Mitarai, K. Komatsu, T. Okimoto, T. Hiratani, H. Yagi, and N. Fujiwara, "Ultrasmall four-channel multiplexer designed by Bayesian direct binary search method", *IEEE Photonics Technology Letters*, vol. 38, no. 11, pp. 703–706, Jun. 2026, DOI: [10.1109/LPT.2026.3663414](https://doi.org/10.1109/LPT.2026.3663414).
- [12] D. Dai, Y. Tang, and J. E. Bowers, "Mode conversion in tapered submicron silicon ridge optical waveguides", *Optics Express*, vol. 20, no. 12, pp. 13425–13439, Jun. 2012, DOI: [10.1364/OE.20.013425](https://doi.org/10.1364/OE.20.013425).